How-To Library

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Using the Replace Table Rule Action to Implement Row-Level Security

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Abstract

This article describes how to create a security rule set to limit the rows that a database request returns and to mask the returned data.

Supported Versions

- Dynamic Data Masking 9.6.0 or later

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Overview

You can use the Replace Table rule action to remove sensitive rows from the result set when an unauthorized user accesses the database. Include a rule that uses the Define Symbol rule action to provide row-level security and limit which rows the user can view. A masking rule removes sensitive columns from the result set.

For example, the Human Resources Manager in the Sales department of your organization must have access to employee data, such as names, birth dates, and contact information. You want to block access to employee information in other departments and you want to block access to all salary information.

Before you begin this example, you must have a database connection and a Dynamic Data Masking service configured in the Management Console. For information about creating database connections and adding services, see the *Dynamic Data Masking Administrator Guide*.

In this example, you perform the following tasks:

1. Create a connection rule that directs database requests to a security rule set.
2. Create a security rule set.
3. Create a security rule folder that identifies requests to the EMPLOYEES table.
4. Create a Define Symbol rule that defines a symbol for the Sales department identification number.
5. Create a Replace Table rule that filters out rows from the result set that contain information about employees in other departments.
6. Create a masking rule that masks salary information.

Step 1. Create the Connection Rule

Create a connection rule that directs incoming requests to a security rule set.

1. In the Management Console, select the Dynamic Data Masking service node for the database and click **Tree** > **Connection Rules**.

   The **Rule Editor** window appears.

2. In the Rule Editor, click **Action** > **Append Rule**.

   The **Append Rule** window appears.

3. Enter the following connection rule parameters:

   Rule Name

   Enter the name of the rule. The rule name appears in the rule tree. In this example, the connection rule name is DirectToRuleSet.

   Matcher

   Select the All Incoming Connections matcher. The All Incoming Connections Matcher applies the connection rule action to all incoming database requests.

   Action

   Select the Use Rule Set rule action. The Use Rule Set action directs the requests to the security rule set that you define in the **Rule Set Name** field.

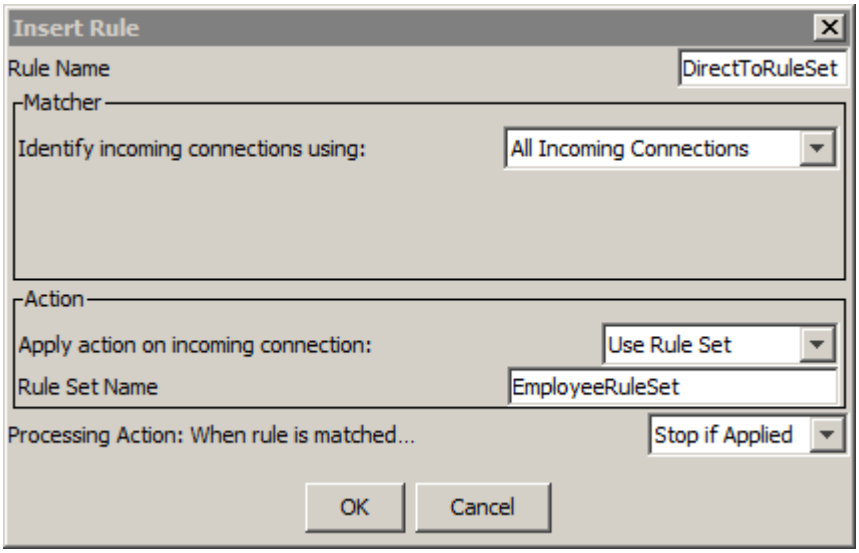
   Rule Set Name

   Enter the name of the security rule set that you want to create to restrict access to the database. In this example, the rule set name is EmployeeRuleSet.

   Processing Action

   Select the Stop if Applied processing action. The Stop if Applied processing action instructs the Rule Engine not to continue to the next rule in the tree.

   The following image shows the connection rule parameters:



4. Note the **Rule Set Name** value that you entered for the connection rule. You use the value to name the security rule set.
5. Click **OK** to add the rule to the rule tree.
6. Click **File** > **Update Rules** to save the connection rule and click **File** > **Exit** to close the Rule Editor.

Step 2. Create the Security Rule Set

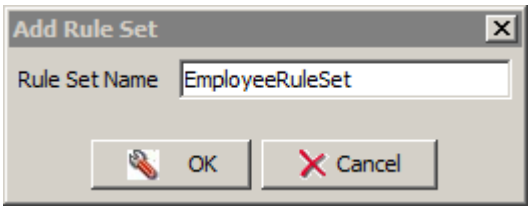
Create the security rule set to hold the security rules.

1. In the Management Console, select the root domain node and click **Tree** > **Add Rule Set**.

   The **Add Rule Set** window appears.

2. In the **Rule Set Name** field, enter the name of the rule set. Enter the same name that you entered in the connection rule **Rule Set Name** field. In this example, the rule set name is EmployeeRuleSet.

   The following image shows the **Add Rule Set** window:



3. Click **OK**.

   The rule set appears in the Management Console tree.

Step 3. Create the Security Rule Folder

Create a security rule folder that identifies requests to the EMPLOYEES table.

1. In the Management Console tree, select the security rule set that you created and click **Tree** > **Security Rule Set**.

   The **Rule Editor** window appears.

2. In the Rule Editor, click **Action** > **Append Rule**.

   The **Append Rule** window appears.

3. Configure the following parameters for the security rule folder:

   Rule Name

   Enter the name of the rule folder. In this example, the rule folder name is Employees.

   Description

   Enter an optional description of the rule folder.

   Matcher

   Select the Text matcher and define the following parameters:

   - **Text**. In the **Text** field, enter a regular expression to match the request. In this example, the regular expression is `.*(EMPLOYEES).*`, which identifies requests to the EMPLOYEES table.
   - **Identification Method**. Select the Regular Expression identification method.

   Action

   Select the Folder action. The Folder action makes the rule a folder into which you can add additional security rules.

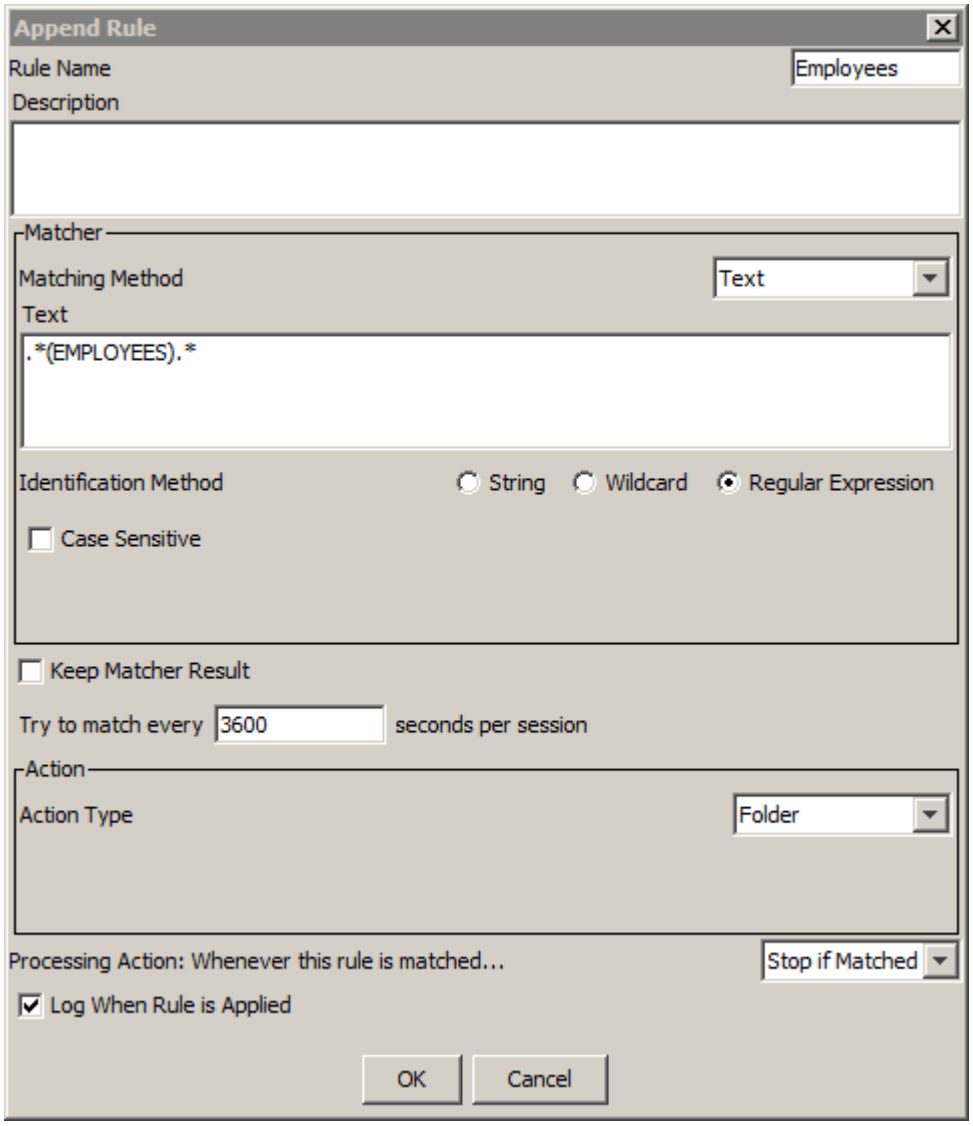
   Processing Action

   Select the Stop if Matched processing action.

Log When Rule is Applied

To include information in the log files about when the Rule Engine applies the rule, select the Log When Rule is Applied parameter.

The following image shows the rule folder parameters:



4. Click **OK** to close the close the window and add the rule to the rule tree.
5. Click **File** > **Update Rules** to save the rule.

Step 4. Create the Define Symbol Rule

Create a Define Symbol rule that defines a symbol for the Sales department identification number.

1. In the Rule Editor, select the security rule folder that you created and click **Action** > **Append Rule**.

   The **Append Rule** window appears.

2. Configure the following parameters for the Define Symbol rule:

Rule Name

Enter the name of the rule. In this example, the rule name is DefineSymbol.

Description

Enter an optional description of the rule.

Matcher

Select the Any matcher. The Rule Engine applies the rule to all incoming requests.

Action

Select the Define Symbol action. In the Symbol Definition box, define the following parameters:

- **Symbol Name**. The name of the symbol. In this example, the symbol name is DepartmentID.
- **Symbol Value**. The value of the symbol. In this example, the symbol value is 100, which is the Sales department identification number in the DEPARTMENT_ID column of the EMPLOYEES table.
- **Keep per Session**. Enter Yes. The symbol becomes a global symbol for the session and exists until the user disconnects from the database.

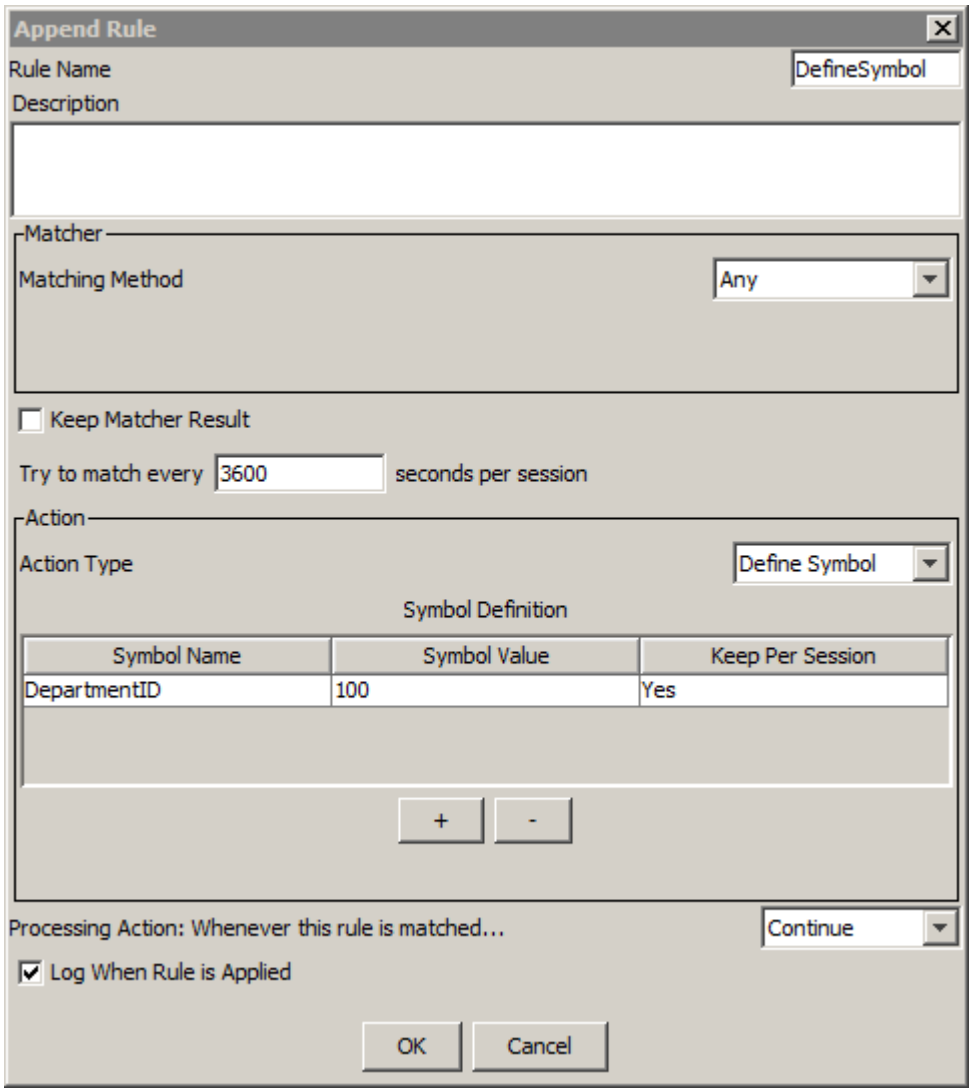
Processing Action

Select the Continue processing action. The Rule Engine continues to the next rule in the tree.

Log When Rule is Applied

To include information in the log files about when the Rule Engine applies the rule, select the Log When Rule is Applied parameter.

The following image shows the rule parameters:



3. Click **OK** to close the window and add the rule to the rule tree.
4. Click **File** > **Update Rules** to save the rule.

Step 5. Create the Replace Table Rule

Use the symbol that you defined in Step 4 to create a Replace Table rule that filters out rows from the result set that contain information about employees in other departments.

1. In the Rule Editor, select the security rule folder that you created and click **Action** > **Append Rule**.

   The **Append Rule** window appears.
2. Configure the following parameters for the Replace Table rule:

   Rule Name

   Enter the name of the rule. In this example, the rule name is ReplaceTable.

Description

Enter an optional description of the rule.

Matcher

Select the Any matcher. The Rule Engine applies the rule to all incoming requests.

Action

Select the Replace Table action. The Replace Table action replaces the FROM clause of the SQL statement to provide row-level security. In the Tables to Replace box, define the following parameters:

- **Table Name**. Enter the name of the table. In this example, the table name is `.*EMPLOYEES.*`, which identifies the EMPLOYEES table.
- **Alternative Object or Query**. Enter an alternative query that includes a WHERE clause to replace the original FROM clause in the request. In this example, the alternative query is `SELECT * FROM \(table) WHERE DEPARTMENT_ID=\(DepartmentID)`. The alternative query changes the statement so that it returns only rows that have the Sales department identification number in the DEPARTMENT_ID column. The query specifies the Sales department identification number with the DepartmentID symbol.

  Note: You can use the \(table), \(prefix), and \(tableOnly) symbols with the Replace Table action to specify the part of the FROM clause that you want to replace. For more information about the Replace Table action symbols, see the *Dynamic Data Masking User Guide*.

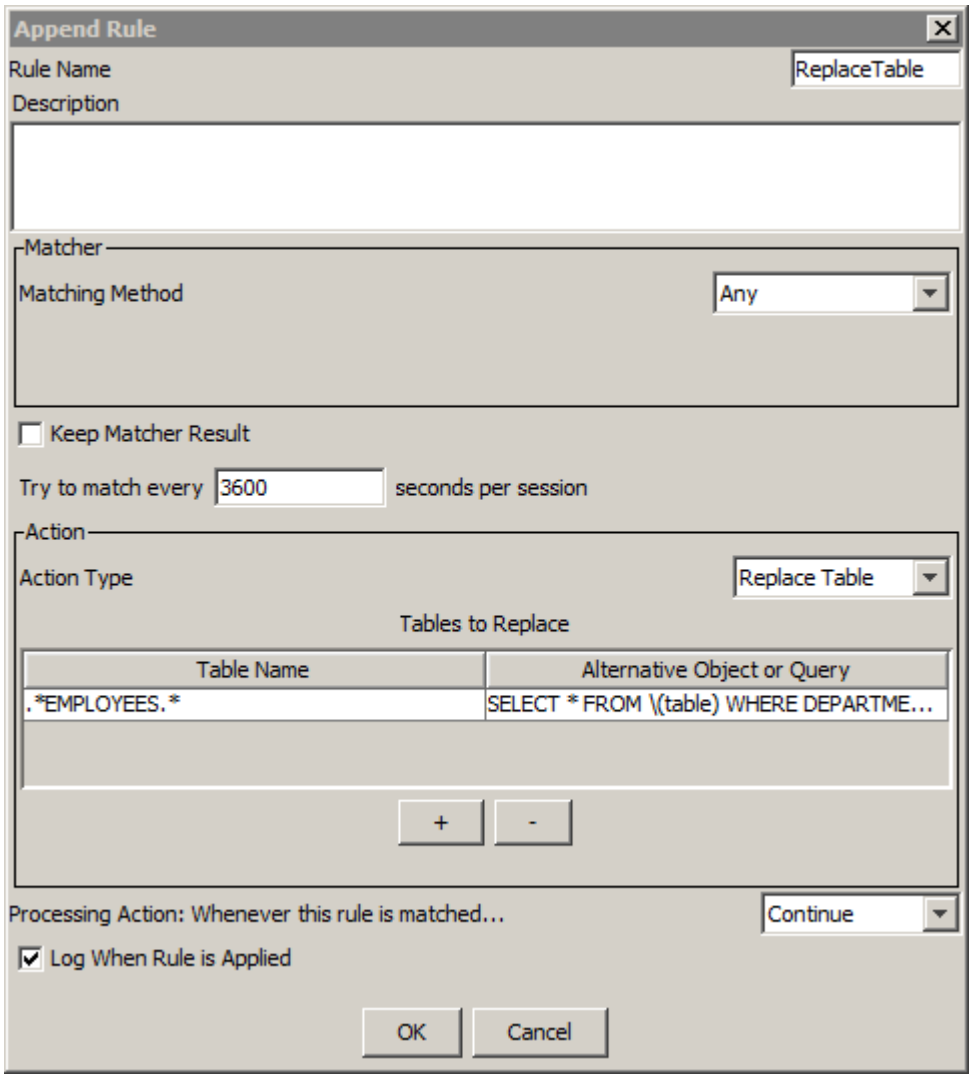
Processing Action

Select the Continue processing action. The Rule Engine continues to the next rule in the tree.

Log When Rule is Applied

To include information in the log files about when the Rule Engine applies the rule, select the Log When Rule is Applied parameter.

The following image shows the rule parameters:



3. Click **OK** to close the window and add the rule to the rule tree.
4. Click **File** > **Update Rules** to save the rule.

Step 6. Create the Masking Rule

Create a masking rule that masks salary information.

1. In the Rule Editor, select the security rule folder that you created and click **Action** > **Append Rule**.
   The **Append Rule** window appears.
2. Configure the following parameters for the masking rule:

   Rule Name

   Enter the name of the masking rule. In this example, the rule name is Mask.

Description

Enter an optional description of the rule.

Matcher

Select the Any matcher. The Rule Engine applies the rule to all incoming requests.

Action

Select the Mask action. When you select the Mask action, the Rule Engine applies the masking function that you define. Click the plus sign (+) to add a masking function. In the Columns to Mask box, define the following parameters:

- **Table Name**. Enter the name of the table. In this example, the table name is `.*EMPLOYEES.*`, which identifies the EMPLOYEES table.
- **Column Name**. Enter the name of the column. In this example, the table name is `.*SALARY.*`, which identifies the SALARY table.
- **Masking Function**. Enter a masking function to mask the sensitive data in the column. In this example, the masking function is `substr(\(col),'')`, which returns null values in the SALARY column.

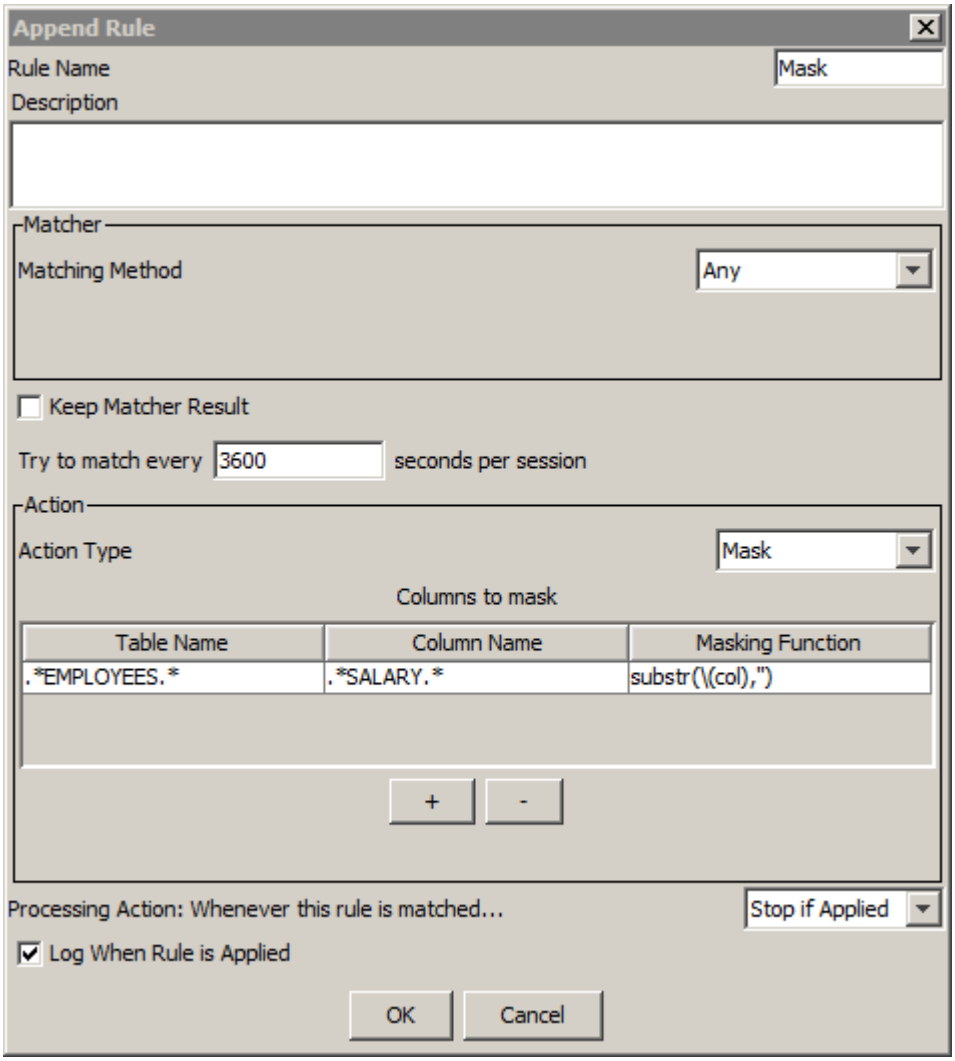
Processing Action

Select the Stop if Applied processing action. If the Rule Engine applies the masking function, the Rule Engine does not continue to the next rule in the tree.

Log When Rule is Applied

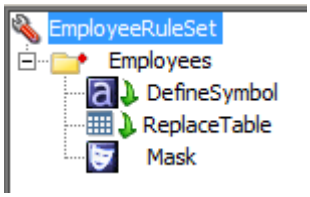
To include information in the log files about when the Rule Engine applies the rule, select the Log When Rule is Applied parameter.

The following image shows the rule parameters:



3. Click **OK** to close the window and add the rule to the rule tree.
4. Click **File** > **Update Rules** to save the rule.

The following image shows the security rule set tree with the rule folder and security rules:

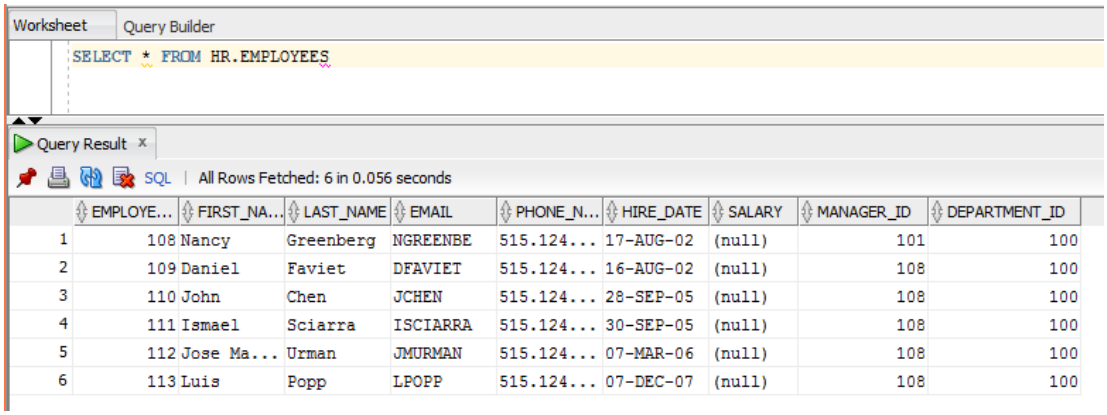


Test the Rule Set

Verify that you created the connection and security rules correctly.

After you create the rule set, you can use the Security Rule Set Simulator to verify that the rules correctly restrict and mask data. To use the simulator, select the root security rule set node and click **Action** > **Security Rule Set Simulator**. For more information about the Security Rule Set Simulator, see the *Dynamic Data Masking User Guide*.

You can run a query on the database to view the result set. The following image shows the result set of a query from SQL Developer:



	EMPLOYE...	FIRST_NA...	LAST_NAME	EMAIL	PHONE_N...	HIRE_DATE	SALARY	MANAGER_ID	DEPARTMENT_ID
1	108	Nancy	Greenberg	NGREENBE	515.124...	17-AUG-02	(null)	101	100
2	109	Daniel	Faviet	DFAVIET	515.124...	16-AUG-02	(null)	108	100
3	110	John	Chen	JCHEN	515.124...	28-SEP-05	(null)	108	100
4	111	Ismael	Sciarra	ISCIARRA	515.124...	30-SEP-05	(null)	108	100
5	112	Jose Ma...	Urman	JMURMAN	515.124...	07-MAR-06	(null)	108	100
6	113	Luis	Popp	LPOPP	515.124...	07-DEC-07	(null)	108	100

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Acknowledgements

Thanks to the Dynamic Data Masking development and QA teams for their help in completing this article.